

MARVIN ISHQ YONAN

Electronics & Embedded Systems Engineer

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PROFILE

Graduate engineer from INSA Toulouse (Critical Embedded Systems — SIEC), with hands-on experience across the full embedded stack: hardware design (analog circuits, PCB layout, KiCad/LTSpice), digital design (VHDL/FPGA), low-level firmware (C on STM32/ESP32), communication protocols (SPI, I2C, UART, LoRa, NFC), hardware security (ESE/HSM, ISO 26262), and test infrastructure (CI/CT, Jenkins). Background in safety-critical development standards (ISO 26262, DO-178/ED-12) and AUTOSAR. Available from September 2026.

PROFESSIONAL EXPERIENCE

Aumovio · Toulouse, France (formerly Continental Automotive)

Feb 2026 – Present

Embedded Systems Engineering Intern

- **Safety-critical automotive embedded development** on a production program integrating hardware security elements (ESE/HSM) and hardware cryptographic mechanisms (MAC/AES), under strict ISO 26262 functional safety constraints.
- **Root-cause analysis and bug resolution** on SPI T=1 communication between MCUs and secure elements, identifying protocol-level faults, reproducing defects on testbenches, and delivering fixes validated before client delivery.
- **Secure flash tooling**, developed and maintained Python/Java utilities for authenticated programming of embedded security components across multiple ECU configurations.
- **Requirements & interface traceability**, maintained hardware/software interface documentation in IBM DOORS across electronic components, ensuring consistency between specs and implementation.
- **CI/CT pipeline integration (in progress)**: building an automated test infrastructure using Jenkins, Bash, and Python to run unit and integration tests on physical testbenches, targeting repeatable, regression-proof validation before each client shipment.
- **Internal AUTOSAR training**, completed internal modules covering AUTOSAR layered architecture (RTE, BSW, MCAL) in the context of automotive ECU software integration.
- **International Agile collaboration** with cross-functional teams in Singapore and Germany via Jira and GitHub, in a structured sprint-based development cycle.

Phoenix Hellas · Greece

Jul 2025 – Sep 2025

Industrial Automation & Robotics Engineering Intern

- **Robotic cell design and commissioning** — programmed and validated collaborative robot sequences using RoboDK and OEM simulation platforms; optimised trajectories and peripheral coordination for production lines.
- **PLC programming** — designed and commissioned ladder/structured-text programs for industrial automation lines, including servo synchronisation for improved motion precision.

EDUCATION

INSA Toulouse · France

Diplôme d'Ingénieur — Electronic and Control Systems Engineering, Critical Embedded Systems (SIEC)

2021 – 2026

Equivalent to an accredited Master of Engineering (CTI-accredited, EUR-ACE recognised, Grande École status).

- **Core curriculum**: Analog & digital electronics, signal processing (filtering, spectral analysis), power electronics, control systems, real-time OS (RTOS), microcontroller architecture.
- **Safety-critical track (SIEC)**: Functional safety concepts (DO-178/ED-12, ISO 26262 principles), formal methods (SCADE/Lustre), dependability (FMEA, fault trees), certifiable software development lifecycle.
- **Hardware design**: VHDL / FPGA design, PCB schematic capture and layout (KiCad), simulation (LTSpice, MATLAB/Simulink), ADA programming for high-integrity systems.

Collège Notre Dame de Louaizé · Lebanon

Scientific Baccalaureate, Highest Honours (Mention très bien)

2018 – 2021

ENGINEERING PROJECTS

RISC-V-like Processor Core in VHDL *VHDL · FPGA · Digital Architecture*

- **First VHDL project:** designed and implemented a RISC-V-inspired pipelined processor from scratch in VHDL: ALU, register file, instruction fetch/decode/execute stages. Synthesised and tested on FPGA target.
- Solidified understanding of digital micro-architecture, clock domains, synthesis constraints, and hardware-software co-design principles.

Thales RISC-V Performance Challenge (unofficial participation) *RISC-V · CVA6 · HPC · FPGA*

- **Frequency optimisation of a RISC-V CVA6 core:** participated in the Thales RISC-V contest aiming to push the maximum operating frequency of the open-source CVA6 core while maintaining correct execution of CoreMark.
- Achieved measurable frequency gains through timing-path analysis and targeted RTL optimisation. Informal participation due to team registration issues, full technical work completed.

Autonomous Sailboat — Embedded Control System *STM32 · C · SPI · I2C · GPIO · PWM · Timers*

- **Primary MCU project:** developed the full embedded software stack on an STM32 to control a sailboat autonomously: real-time sail actuation via PWM-driven servomotors, incremental encoder reading, wind-sensor data acquisition.
- Implemented and debugged SPI and I2C peripheral drivers from scratch using STM32 HAL (timers, DMA, interrupt handlers, GPIO), gaining deep familiarity with bare-metal embedded C development.

Electric Scooter — Analog Design, Filtering & PCB *LTSpice · KiCad · Analog · Power Electronics*

- **Full analog signal-chain design:** dimensioned and simulated a Butterworth filter (LTSpice) for motor current signal conditioning, then transferred the design to KiCad for PCB schematic capture, component placement and routing.
- Designed regenerative braking circuitry to return kinetic energy to the battery pack, applying power electronics principles to a real electromechanical system.

LoRa & NFC Banking PoC — ESP32 IoT System *ESP32 · LoRa · NFC · C · IoT Protocols*

- **Personal full-stack IoT project:** built a proof-of-concept distributed banking system comprising a server node (the 'bank') and a POS terminal, communicating over LoRa radio links with NFC card authentication, all running on ESP32 microcontrollers.
- Designed the complete protocol stack from scratch: LoRa frame format, NFC UID-based authentication flow, and lightweight transaction state machine. Reinforced knowledge of wireless link budgets, range estimation, and spread-spectrum modulation.

Connected Object Design — From Sensor to Antenna (OpenClassrooms / INSA Toulouse MOOC) *RF · Antenna · ADC · PCB · Energy Harvesting*

- **40-hour self-study course** by INSA Toulouse professors (LAAS-CNRS): analog sensor signal chains (ADC theory, protocol configuration), analog front-end dimensioning (transfer function, op-amp stages), wireless link budget, antenna selection and matching, battery charging architectures (wired & wireless), and full PCB design flow from schematic to routed board.

TECHNICAL SKILLS

Languages	C / C++ · Python · VHDL · ADA · Lustre/SCADE · Java · Assembly · Bash · LaTeX
Microcontrollers	STM32 (HAL, bare-metal) · ESP32 · Keil µVision · PSOC Programmer
Protocols	SPI · I2C · UART · SPI T=1 · LoRa · NFC · CAN (awareness) · ARINC 429 (awareness)
HW Design	KiCad (schematic + PCB layout) · LTSpice · Analog filter design · Power electronics · RF link budget · Antenna matching
Safety Standards	ISO 26262 · DO-178B/C / ED-12 · AUTOSAR layered architecture · SCADE/Lustre · FMEA basics
Simulation	MATLAB / Simulink · LTSpice · RoboDK · Creo
DevOps / Tools	Jenkins · Git / GitHub · Jira · IBM DOORS · Visual Studio / VSCode

LANGUAGES

French Native (C2) · **English** Fluent (C2) · **Arabic** Native (C2) · **Spanish** Professional (B2)